



SENIOR DESIGN 2026

VibeSit Insight

A Rehabilitation Platform Redesign for Vibration Therapy

Sabrina Simkhovich · Mitchell Salzman · Hunter Johnson · Johannes Winarta | Advisors: Jeff Leismer & Ahmet Turkmen



UNIVERSITY OF MINNESOTA
Electrical & Computer Engineering

CLIENT · VIBETECH

INTRODUCTION

• ABSTRACT

VibeTeam redesigned the companion iPad application for the VibeTech VT3 into VibeSit Insight, a Swift/SwiftUI platform that guides patients and clinicians through a complete rehabilitation program. The application delivers adaptive check-in surveys, standardized clinical tests for balance, mobility, and strength, a composite scoring system, and a progress dashboard.

• BACKGROUND

Lower-limb rehabilitation is essential for patients recovering from surgery, managing chronic conditions, or facing age-related declines in strength and mobility, yet existing tools rarely provide the infrastructure to objectively track outcomes over time. The VibeTech VT3 device delivers resistance and vibration therapy while measuring force and repetitions, but its original software lacked a structured workflow and centralized data collection, leaving progress scattered across paper forms, spreadsheets, and clinician notes. A redesigned application was needed to bridge the gap between a capable physical therapy device and the clinical workflows required to make its data actionable.

• OBJECTIVES

- ◆ Intuitive UI/UX for elderly users
- ◆ Treatment infrastructure
- ◆ Adaptive pre- and post-session surveys
- ◆ Progress dashboard with trend visualization
- ◆ Add Home Screen, User guide and Profile
- ◆ Implement data collection



Figure 1. Device in use

METHODOLOGY / DESIGN OVERVIEW

• SYSTEM ARCHITECTURE DESIGN

The VibeSit Insight software stack connects a SwiftUI iPad application to a Python Flask REST API via a secure ngrok tunnel, transmitting authenticated user survey, session, and assessment data using HTTPS and JWT Bearer tokens. All patient records, treatment configurations, and progress metrics are persisted in a MongoDB database, which the app queries in real time to populate the Dashboard and Profile pages throughout the treatment lifecycle.

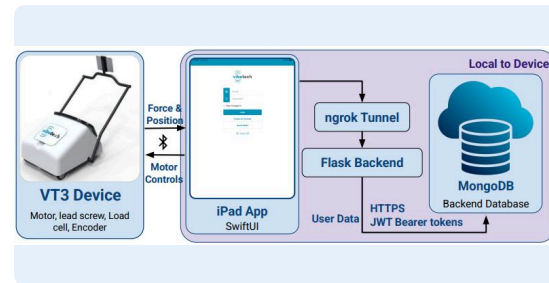


Figure 2. System architecture

App flow is organized into five functional phases that define the complete patient journey from onboarding through completing treatment.

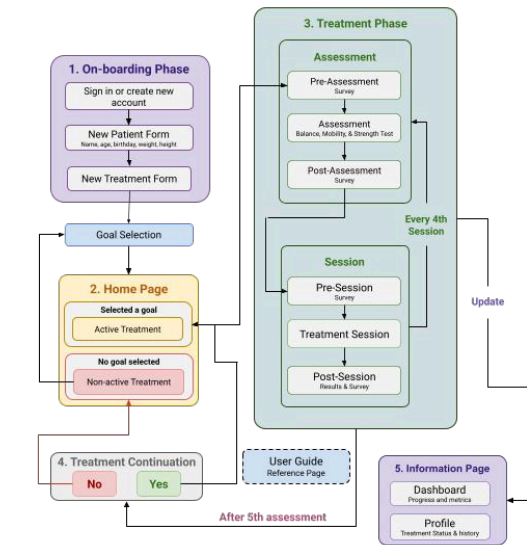


Figure 3. App flow diagram

METHODOLOGY, CONT.

• IMPLEMENTATION

VibeSit Insight was built in SwiftUI, with a MongoDB backend accessed via an authenticated REST API. We used a cloud macOS environment (RentAMac) was provisioned to give all developers remote access to Xcode where code was simulated on a virtual iPad.

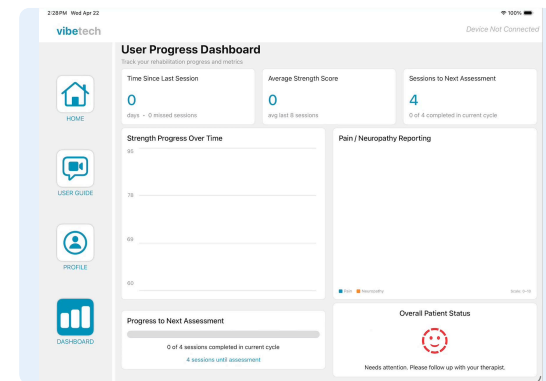


Figure 4. Patient dashboard with session history and strength trend chart.

• TESTING & VALIDATION

The system's functionality was tested by simulating therapy sessions to ensure proper workflow performance and data accuracy. Each data point entered through the app was verified to have a 1:1 match in the MongoDB backend, confirming that every survey response, assessment result, and session metric was captured, stored, and retrievable as expected. The testing proved that treatment plans and assessments were scheduled appropriately, session information was correctly recorded, and data reports were consistently generated from the dashboard.

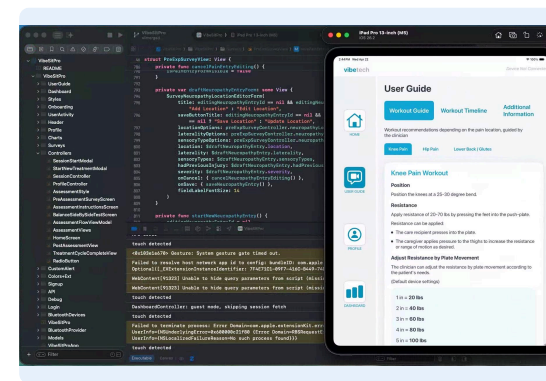


Figure 5. Xcode development environment with simulator running our code.

RESULTS

• BEFORE & AFTER COMPARISON

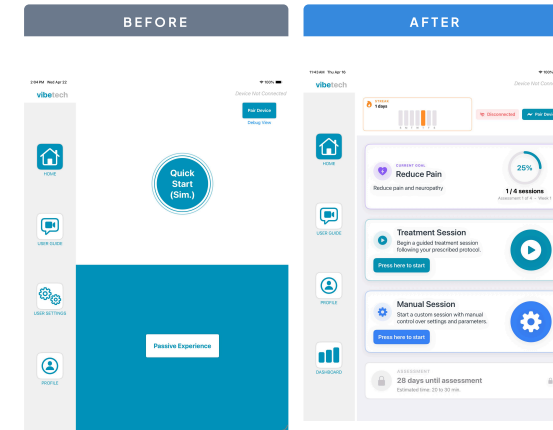


Figure 5. Legacy application screen.

Figure 7. Redesigned VibeSit Insight interface.

CONCLUSION

• IMPACT

VibeSit Insight transforms the VT3 from a standalone therapy device into a complete clinical platform.

Patients gain an accessible interface designed for elderly users, with adaptive surveys that reduce repetitive data entry, session-by-session progress visualization, and a dashboard that makes their rehabilitation journey tangible and motivating.

Clinicians gain standardized, SPPB-aligned assessments, configurable treatment timelines, and measurable longitudinal outcomes, replacing spreadsheets with a centralized, objective data pipeline that supports confident treatment decisions.

• FUTURE WORK

- ◆ Integrate VT3 hardware
- ◆ AI predictive models to progress trends
- ◆ Implement adaptive treatment
- ◆ Extend platform support Android

• ACKNOWLEDGEMENTS

The team would like to thank **Jeff Leismer** (VibeTech) for external advisory support, **Bryan Doss** (EE Solutions) for technical help, **Prof. Ahmet Turkmen** for faculty supervision, and **Dave Ballard** for purchasing assistance.